



Inflammation

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Pathology 202



Cardinal signs of inflammation

Rubor (redness)

Tumor (swelling)

Calor (heat)

Dolor (pain)

Functio laesa (loss of function)



Vascular changes

Vasodilation is the earliest vascular change in acute inflammation.

Increased vascular permeability leads to exudation of protein-rich fluid.

Stasis of blood flow allows neutrophils to marginate along the endothelium.



Cellular events

Neutrophils are the first cells to arrive in acute inflammation.

Neutrophils are replaced by monocytes after 24 to 48 hours.

Selectins mediate rolling and integrins mediate firm adhesion of leukocytes.

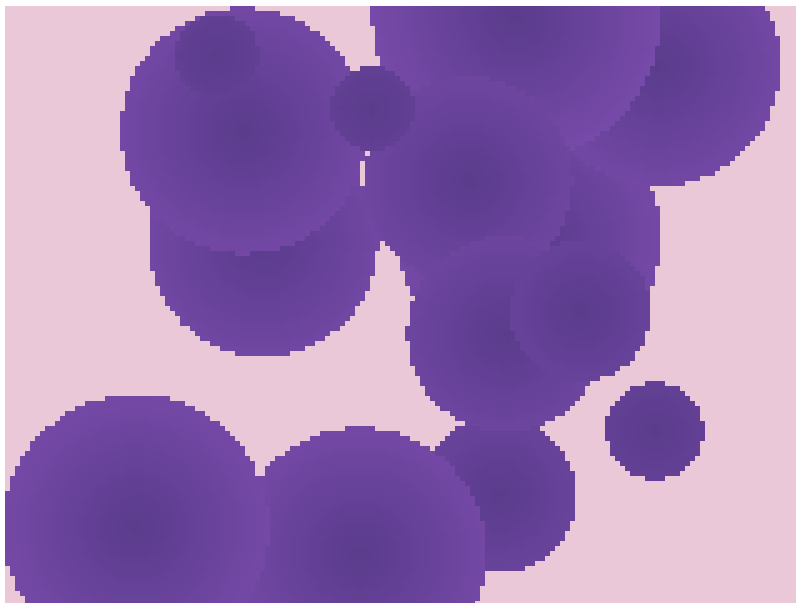


Figure 4.1: neutrophils in an alveolar exudate



Chemical mediators

Histamine causes arteriolar dilation and increased venular permeability.

Prostaglandins mediate pain and fever.

Bradykinin causes pain and increased vascular permeability.



References

Robbins and Cotran Pathologic Basis of Disease, 10th edition.

Thank you